

1. INTRODUCTION

In today's world, automation plays a crucial role in improving convenience, security, and efficiency in daily life. With the rapid advancement of technology, automated systems are increasingly being adopted in both residential and commercial environments. One such important application of automation is the automatic door opening system. This system is widely used in places such as shopping malls, hospitals, airports, offices, and smart homes, where manual operation of doors can be inconvenient, time-consuming, or unhygienic.

The automatic door opening system is designed to open and close doors automatically by detecting the presence of a person or receiving an input signal. This is achieved using sensors and a microcontroller-based control system. In this project, the 8051 microcontroller is used as the main processing unit, which controls the operation of the door based on input signals received from sensors or switches.

The system operates by detecting a signal (such as from a push button, infrared sensor, or motion detector), after which the microcontroller processes the signal and generates appropriate control outputs. These outputs are then applied to a motor driver circuit, which drives a DC motor. The motor rotates in clockwise or anticlockwise direction to open or close the door accordingly. Proper timing delays are implemented in the program to ensure smooth operation and prevent sudden movements.

In addition, the system incorporates basic control logic to avoid unnecessary operation and to ensure that the door opens and closes only under required conditions. This makes the system more reliable and energy-efficient. The use of a motor driver also protects the microcontroller from high current damage, ensuring safe and stable performance.

Moreover, such systems improve safety, accessibility, and hygiene by minimizing physical contact, which is especially important in public places and healthcare environments. They are also beneficial for elderly and disabled individuals, providing ease of access without manual effort.

This project demonstrates the practical implementation of embedded systems and highlights the use of the 8051 microcontroller in real-world automation applications. It also provides a foundation for further enhancements such as sensor-based automation, remote control operation, and integration with smart home systems.

1.1 Need

In modern times, there is a growing demand for automation in daily life to improve convenience, efficiency, and safety. Manual operation of doors in places such as hospitals, shopping malls, offices, and public buildings can be inconvenient, time-consuming, and sometimes unhygienic. Therefore, there is a need for an automatic door opening system.

The automatic door system reduces the need for human effort by automatically opening and closing the door when required. This is especially useful in places with high human traffic, where manual door handling can cause delays and discomfort.

Another important need of this project is to improve hygiene and safety. In public places, many people touch door handles, which can spread germs and infections. An automatic system minimizes physical contact, making it more suitable for hospitals and healthcare environments.

Additionally, this project is important from an educational perspective. It helps in understanding the practical implementation of embedded systems, microcontroller programming, and hardware interfacing. It also provides knowledge about motor control and automation techniques.

The system is cost-effective, reliable, and can be easily implemented and modified for future advancements such as sensor-based control and smart home integration.

Thus, the need for this project arises from the increasing demand for automation, safety, efficiency, and modern technological solutions in everyday applications

1.2 Objective

The primary objective of this project is to design and implement an automatic door opening and closing system using the 8051 microcontroller, integrated with an infrared (IR) sensor and a motor driver circuit. The system is developed to automate the operation of doors by detecting the presence of a person and controlling the movement of the door without any manual effort. This project aims to demonstrate the practical application of embedded systems by combining hardware components such as sensors, microcontrollers, and actuators into a single efficient system.

In addition, the project focuses on improving convenience, safety, and hygiene in modern environments such as hospitals, offices, and public places. By minimizing human contact and ensuring smooth operation, the system provides an effective solution for automated access control. The implementation also helps in understanding real-time system behaviour, signal processing, and decision-making using a microcontroller. Furthermore, the system is designed to be cost-effective, reliable, and easily extendable for future enhancements like smart automation and security integration.

❖ The specific objectives of the project are as follows:

- To design and develop an automatic door control system using the 8051 microcontroller as the central processing unit.
- To interface an infrared (IR) sensor for detecting the presence or movement of a person near the door.
- To implement a control algorithm that processes sensor input and generates appropriate output signals.
- To control a DC motor using the L293D motor driver IC, ensuring safe interfacing between the microcontroller and the motor.
- To achieve bidirectional rotation of the motor for opening and closing the door effectively
- To incorporate proper timing delays for smooth, stable, and controlled door movement.
- To ensure reliable and continuous operation of the system under real-time conditions.
- To reduce human effort and improve user convenience through automation.

- To enhance safety, security, and hygiene by minimizing direct physical interaction with door systems.
 - To analyse system performance in terms of efficiency, response time, and power utilization.
 - To gain hands-on experience in embedded system design, programming, circuit implementation, and hardware-software integration.
 - To develop a scalable system that can be further upgraded with advanced technologies such as motion sensors, RFID-based access control, and IOT-based smart systems.
-
- To automate door operation without requiring manual intervention.
 - To detect the presence of a person using an Infrared (IR) sensor.
 - To control the opening and closing of the door automatically based on sensor input.
 - To interface the IR sensor with the microcontroller for real-time object detection.
 - To control the DC motor using a motor driver circuit for smooth door movement.
 - To demonstrate the practical application of embedded systems in automation.
 - To integrate hardware components such as sensors, actuators, and controllers into a single system.

1.3 Proposed Work

The proposed work of this project is to design and implement an intelligent automatic door opening and closing system using the 8051 microcontroller, integrated with an infrared (IR) sensor and a motor driver circuit. The system is aimed at automating door operations by detecting the presence of a person and performing the required action without manual intervention. This work focuses on developing a reliable and efficient embedded system that can be used in real-time applications.

In the proposed system, the IR sensor continuously monitors the surrounding area to detect the presence or movement of a person near the door. When an object is detected, the sensor generates a signal which is fed as input to the 8051 microcontroller. The microcontroller processes this input based on the programmed logic and decides whether to open or close the door.

The control signals generated by the microcontroller are then applied to the motor driver IC (L293D), which serves as an interface between the low-power microcontroller and the high-power DC motor. The motor driver ensures proper current amplification and protects the microcontroller from electrical damage. Based on the input conditions, the DC motor rotates in clockwise direction to open the door and in anticlockwise direction to close it.

To ensure smooth and safe operation, appropriate timing delays are incorporated in the program. Once the door is opened, it remains open for a predefined duration and then closes automatically. This prevents abrupt movement and enhances the overall performance of the system.

The proposed work also emphasizes system stability, energy efficiency, and operational reliability. The motor is activated only when necessary, reducing power consumption and mechanical wear. Proper circuit design, grounding, and interfacing techniques are used to ensure consistent performance under different operating conditions.

Furthermore, the system is designed with flexibility for future enhancements. It can be upgraded by integrating additional features such as motion sensors, security access systems (RFID or password-based), alarm systems, or IOT-based remote monitoring and control. This makes the project scalable and suitable for modern smart environments.

2. SYSTEM DESIGN

2.1 Block Diagram

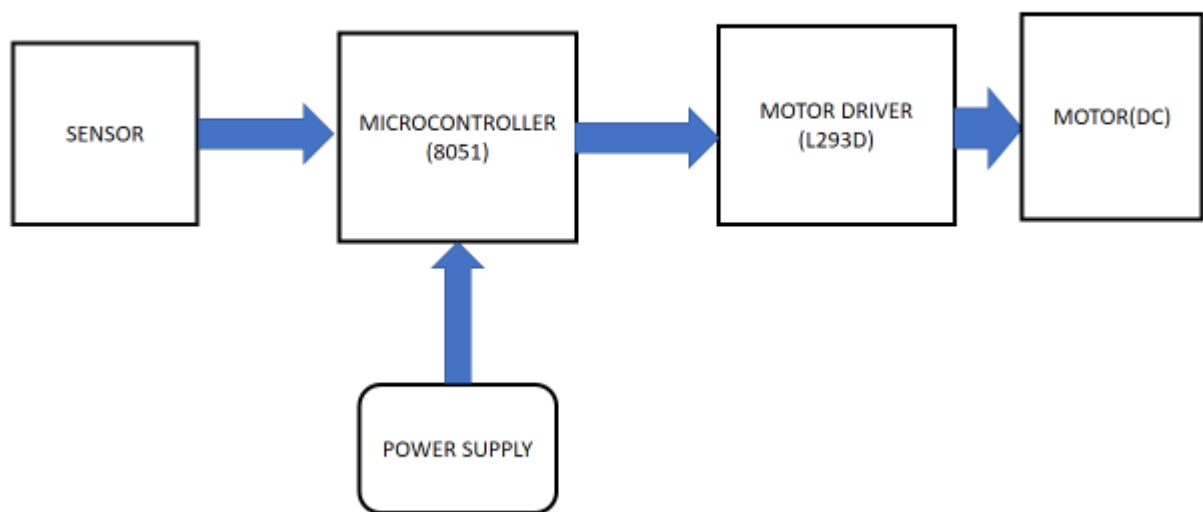


Fig.1 Block Diagram of Automatic Door Opening System

2.2 Circuit Simulation

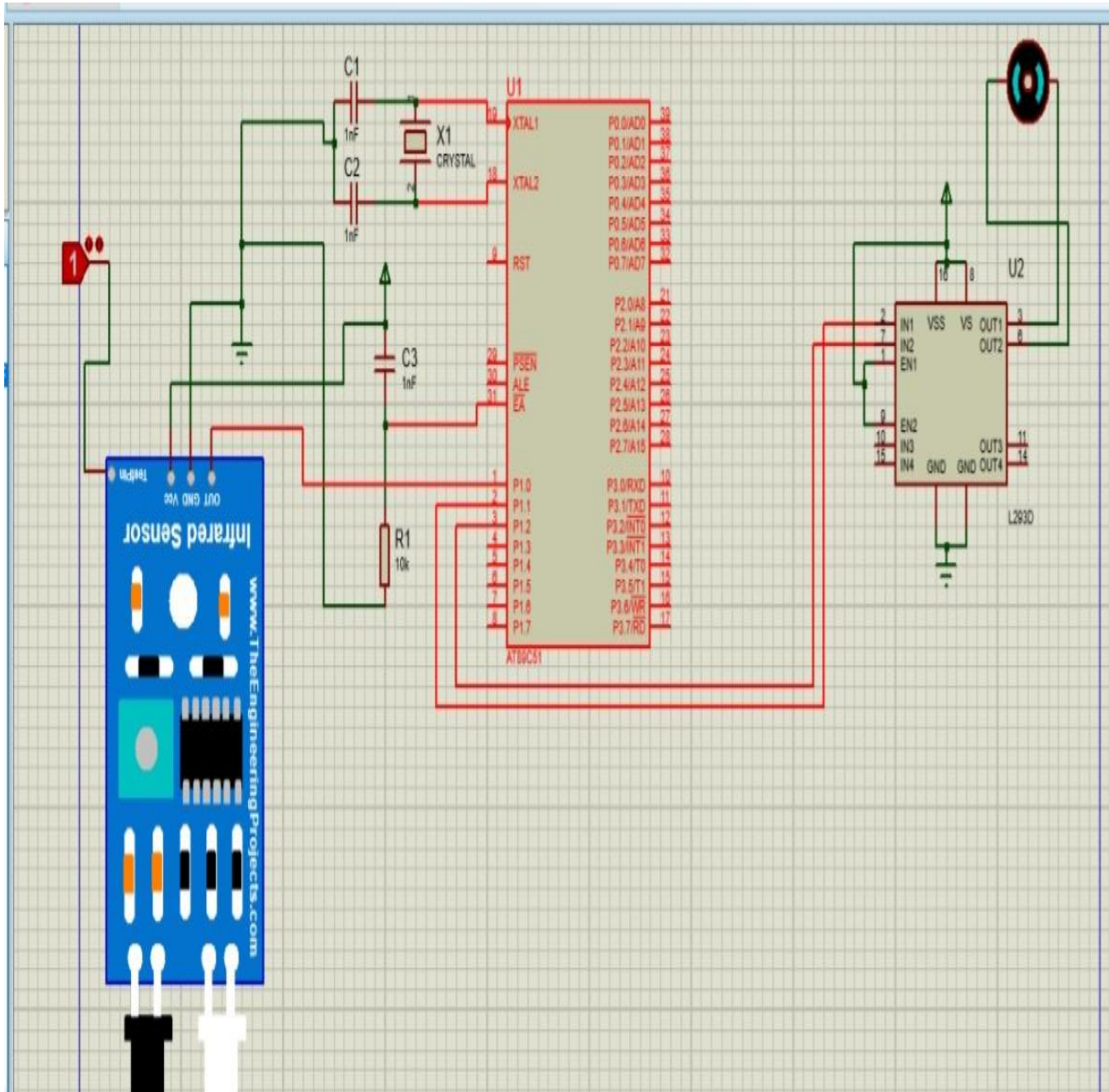


Fig. 2 Circuit Simulation of Automatic Door Opening System

2.3 PCB Layout and Hardware

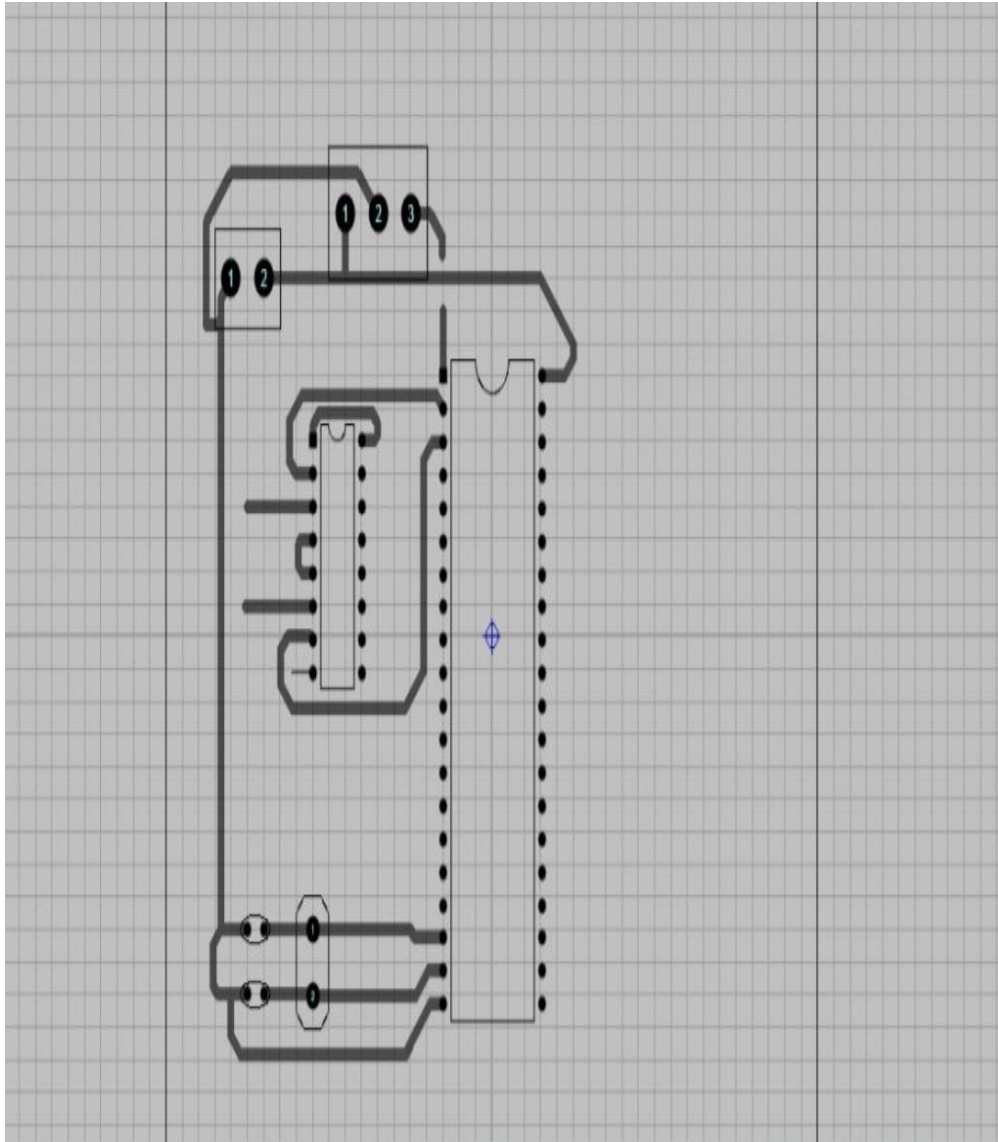


Fig. 3 PCB Layout and Hardware

2.4 Component Description

The automatic door opening system is designed using various electronic components that work together to achieve automation. The description and role of each component used in the project are as follows:

1. Power Supply (9V Battery)

- **Function:**

Provides DC electrical power to the circuit components.

- **Role in the Circuit:**

The 9V battery supplies power to the automatic door opening system. Since the 8051 Microcontroller requires a regulated 5V supply, the battery voltage is reduced and stabilized using a voltage regulator before powering the microcontroller, IR sensor, and other components. This ensures safe and reliable operation of the system.

.



Fig.4 Power Supply (9V Battery)

2. 8051 Microcontroller

- **Function:**

Processes input signals and controls the overall operation of the system.

- **Role-in-Circuit:**

The 8051 microcontroller acts as the brain of the system. It receives signals from the IR sensor, processes them based on the programmed logic, and sends control signals to the motor driver to operate the door.



Fig.5 Microcontroller AT89S51

8051 Microcontroller Pin Description

The 8051 microcontroller is a 40-pin integrated circuit widely used in embedded system applications. Each pin of the microcontroller performs a specific function related to power supply, input/output operations, control signals, and clock generation. The pin description of the 8051 microcontroller is given below:

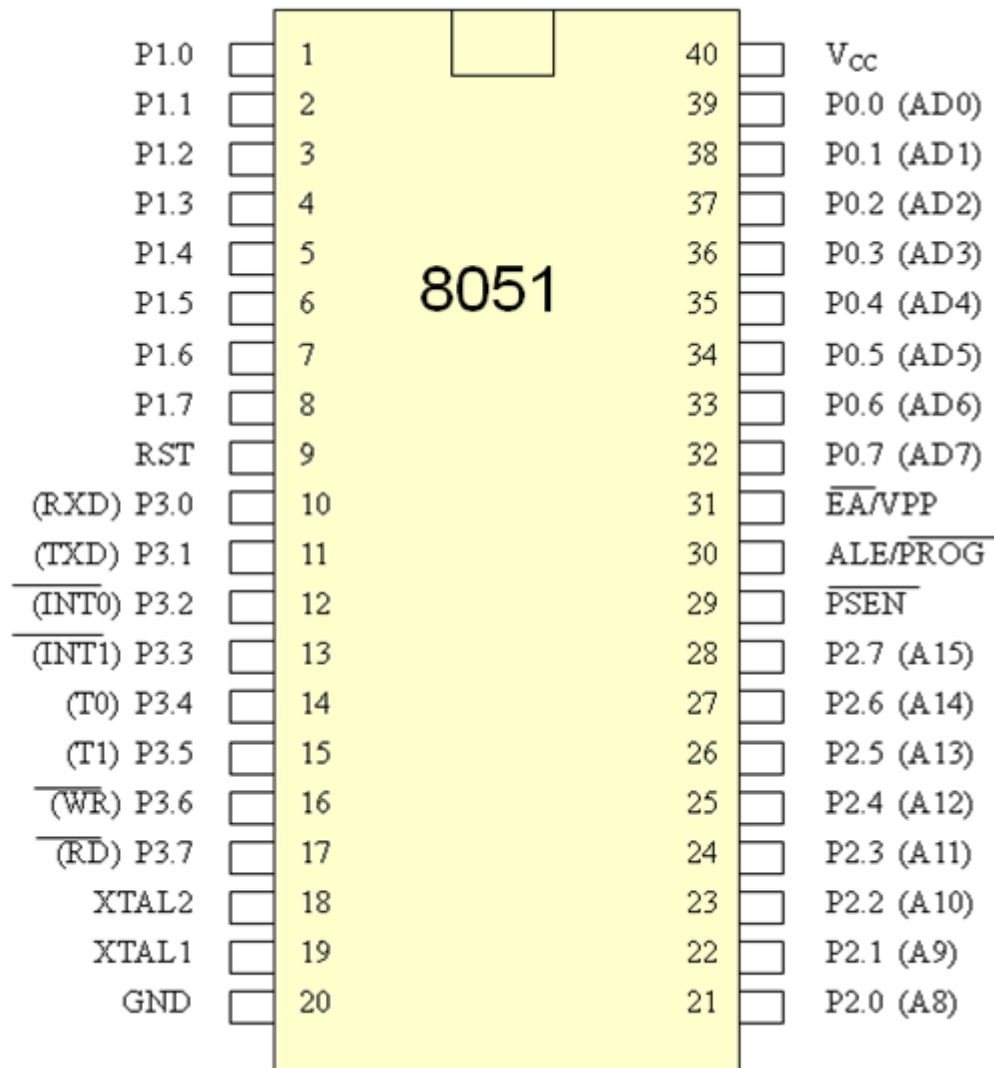


Fig 6. Pin Diagram

1. VCC (Pin 40)

- Provides +5V power supply to the microcontroller.
- Required for the proper operation of the IC.

2. GND (Pin 20)

- Ground pin of the microcontroller.
- Connected to the negative terminal of the power supply.

Port Pins

3. Port 0 (Pins 32–39)

- Consists of pins P0.0 to P0.7.
- Used as a general-purpose input/output port.
- Also functions as a multiplexed address and data bus during external memory interfacing.
- Requires external pull-up resistors.

4. Port 1 (Pins 1–8)

- Consists of pins P1.0 to P1.7.
- Used as a general-purpose input/output port.
- Does not require external pull-up resistors.
- Commonly used for interfacing sensors, LEDs, and switches.

5. Port 2 (Pins 21–28)

- Consists of pins P2.0 to P2.7.
- Used as a general-purpose input/output port.
- Also provides higher-order address bus during external memory interfacing.

6. Port 3 (Pins 10–17)

- Consists of pins P3.0 to P3.7.
- Used as a dual-purpose port with special functions.

Special Functions of Port 3

- **P3.0 (RXD):** Serial data receive pin.
- **P3.1 (TXD):** Serial data transmit pin.
- **P3.2 (INT0):** External interrupt 0.
- **P3.3 (INT1):** External interrupt 1.
- **P3.4 (T0):** Timer 0 external input.
- **P3.5 (T1):** Timer 1 external input.
- **P3.6 (WR):** External memory write control signal.
- **P3.7 (RD):** External memory read control signal.

Control Pins

7. RST (Pin 9)

- Reset pin of the microcontroller.
- Used to restart the program execution.
- A HIGH signal resets the controller.

8. ALE/PROG (Pin 30)

- Address Latch Enable pin.
- Used during external memory interfacing.
- Separates address and data signals from Port 0.

9. PSEN (Pin 29)

- Program Store Enable pin.
- Used for reading external program memory.
- Active LOW signal.

10. EA/VPP (Pin 31)

- External Access pin.
- Selects internal or external program memory.
- Connected to VCC for internal memory operation.

Oscillator Pins

11. XTAL1 (Pin 19)

- Input pin for oscillator circuit.
- Connected to crystal oscillator.

12. XTAL2 (Pin 18)

- Output pin of oscillator circuit.
- Used for clock signal generation.

Internal Data Memory Partitioning:

The internal 256-byte RAM map is split into distinct functional zones based on data grouping and access methods:

1. **Register Banks (00H to 1FH):** Contains 32 bytes mapped to four switchable register banks. Each bank contains eight general-purpose 8-bit working registers (\$R0\$ to \$R7\$). Programmers switch between active banks by manipulating the register selection bits (RS0, RS1) in the Program Status Word (PSW) register.
2. **Bit-Addressable Memory (20H to 2FH):** Provides 16 bytes containing 128 bit-level memory cells. Program instructions can selectively read, clear, or invert singular bits within this range, making it highly efficient for storing binary true/false flags.
3. **General-Purpose Scratchpad (30H to 7FH):** Provides 80 bytes of direct or indirect memory access, typically used for local application variables and runtime CPU stack data.
4. **The Upper Memory Layer (80H to FFH):** This block handles dual operations:
 - **Direct Addressing Access:** Points exclusively to the **Special Function Registers (SFRs)**. The SFR space regulates all internal peripherals, including timer control arrays (TMOD, TCON, T2CON), serial buffer structures (SBUF, SCON), interrupt logic adjustments (IE, IP), and the accumulator (ACC).

AT89S51 information:

The AT89S51 is a low-power, high performance CMOS 8- bit microcontroller with 8k bytes of in-system programmable flash memory. The device is manufactured using Atmel's high-density non-volatile memory technology and is compatible with the industry-standard 80C51 instruction set and pin out.

The Atmel AT89S51 is a powerful microcontroller which provides a highly flexible and cost-effective solution to many embedded control applications.

The AT89S51 Provides the following standard features: 8K bytes of flash, 256 bytes of RAM, 32 I/O lines, Watchdog timer, two data pointers, three 16-bit timer/counters, a six-vector two-level interrupt architecture, a full duplex serial port, on-chip oscillator, and clock circuitry. The AT89C51 is designed with static logic for operation down to zero frequency and supports two software selectable power saving modes.

The power down mode saves the RAM contents but freezes the oscillator, disabling all other chip function until the next interrupt or hardware reset.

The AT89S51 is a low-power, high-performance CMOS 8-bit microcomputer with 8K bytes of Flash programmable and erasable read only memory (PEROM). The device is manufactured using Atmel's high-density non-volatile memory technology and is compatible with the industry-standard 80C51 and 80C52 instruction set and pin out. The on-chip Flash allows the program memory to be reprogrammed in-system or by a conventional non-volatile memory programmer.

By combining a versatile 8-bit CPU with Flash on a monolithic chip, the Atmel AT89C51 is a powerful microcomputer which provides a highly-flexible and cost-effective solution to many embedded control applications.

The AT89S51 provides the following standard features: 8K bytes of Flash, 256 bytes of RAM, 32 I/O lines, three 16-bit timer/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator, and clock circuitry. In addition, the AT89S51 is designed with static logic for operation down to zero frequency and supports two software selectable power saving modes.

The Idle Mode stops the CPU while allowing the RAM, timer/counters, serial port, and interrupt system to continue functioning. The Power-down mode saves the RAM contents but freezes the oscillator, disabling all other chip functions until the next hardware reset.

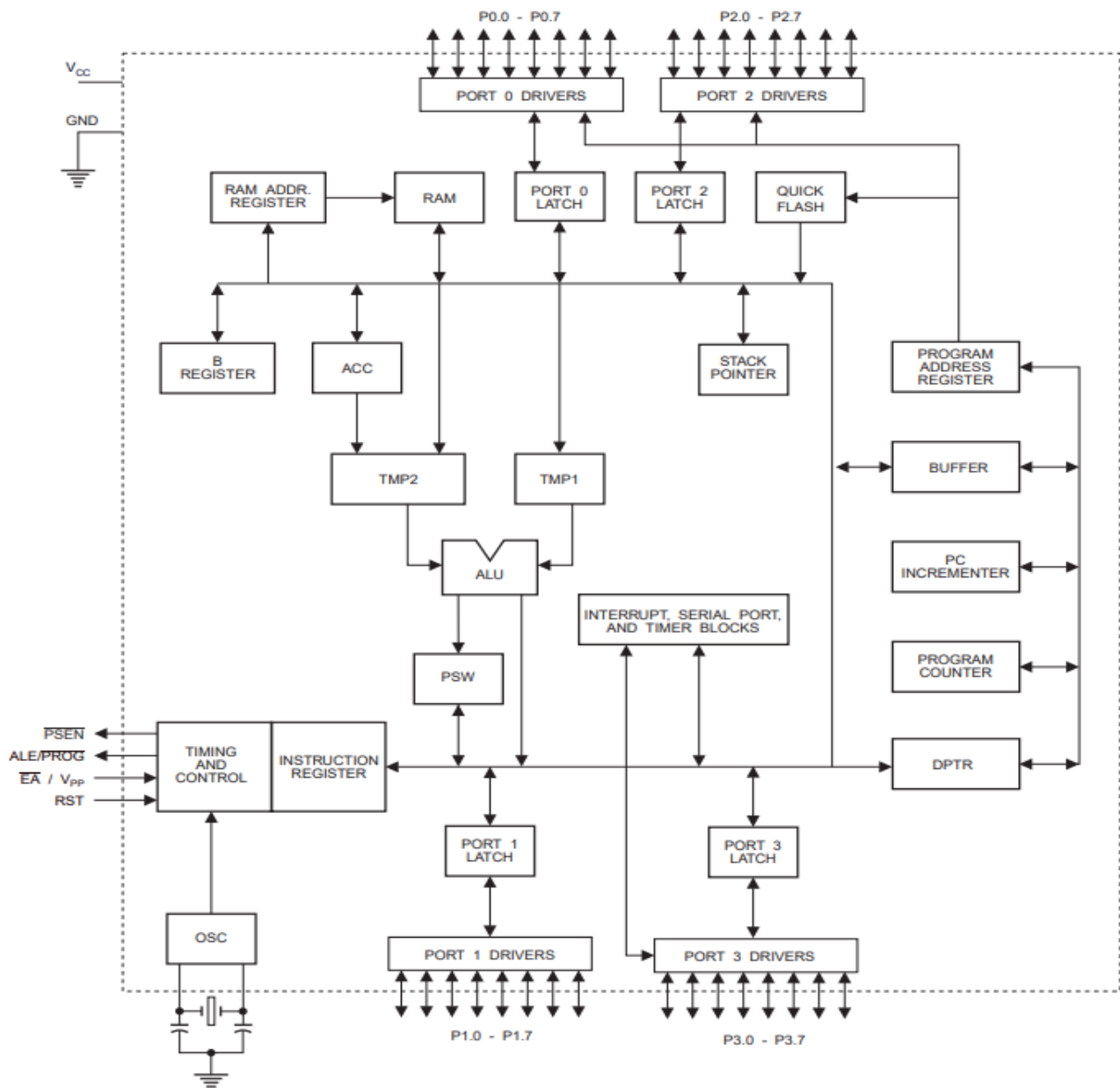


Fig 7. Block Diagram of AT89S51

1. Central Processing Unit (CPU)

Description

The CPU is the main processing unit of the microcontroller. It performs all arithmetic, logical, and control operations. It processes instructions stored in memory and controls the operation of all other internal blocks.

Function

- Executes program instructions.
- Performs arithmetic and logical operations.
- Controls data transfer between internal units.

Role-in-System

In the automatic door system, the CPU processes sensor inputs and controls motor operation according to the programmed logic.

2. Arithmetic Logic Unit (ALU)

Description

The Arithmetic Logic Unit is responsible for performing mathematical and logical operations inside the microcontroller.

Function

- Performs addition and subtraction.
- Executes logical operations such as AND, OR, NOT, XOR.
- Compares data values.

Role-in-System

The ALU helps the microcontroller process sensor signals and make decisions related to opening and closing the door.

3. Accumulator (ACC)

Description

The accumulator is an 8-bit register used for temporary storage of data during arithmetic and logical operations.

Function

- Stores intermediate results.
- Works directly with the ALU during processing.

Role-in-System

It temporarily stores data related to sensor input and control signals during program execution.

4. B Register Description

The B register is another general-purpose register used mainly during multiplication and division operations.

Function

- Assists arithmetic operations.
- Stores temporary data.

Role-in-System

Supports internal calculations performed by the microcontroller.

5. Program Counter (PC) Description

The Program Counter stores the address of the next instruction to be executed.

Function

- Tracks program execution sequence.
- Automatically increments after each instruction.

Role-in-System

Ensures continuous execution of the automatic door control program.

6. Instruction Register Description

The instruction register stores the current instruction being executed by the CPU.

Function

- Holds instruction temporarily during execution.
- Sends instruction to the decoder.

Role-in-System

Helps the microcontroller understand and execute programmed operations correctly.

7. RAM (Random Access Memory)

Description

RAM is temporary memory used for data storage during program execution.

Function

- Stores temporary variables and data.
- Holds intermediate processing information.

Role-in-System

Stores sensor data, timing values, and control variables used in the automatic door system.

8. ROM / Program Memory

Description

ROM stores the program instructions permanently inside the microcontroller.

Function

- Stores embedded C program.
- Retains data even after power OFF.

Role-in-System

Contains the program that controls the automatic opening and closing of the door.

9. Stack Pointer

Description

The stack pointer stores the address of the top of the stack memory.

Function

- Manages stack operations.
- Stores return addresses during function calls.

Role-in-System

Helps in proper execution of subroutines and interrupt handling.

10. Data Pointer (DPTR)

Description

DPTR is a 16-bit register used for addressing external memory.

Function

- Stores external memory addresses.
- Transfers data efficiently.

Role-in-System

Used in advanced applications involving external devices and memory.

11. Port 0, Port 1, Port 2, and Port 3

Description

The 8051 contains four 8-bit input/output ports.

Function

- Interfaces external devices.
- Sends and receives digital signals.

Role-in-System

- Port 1 receives input from the IR sensor.
- Other ports provide output signals to the L293D motor driver and additional devices.

12. Port Drivers and Port Latches

Description

Port drivers strengthen signals sent from ports, while latches temporarily store port data.

Function

- Maintain stable data transmission.
- Improve interfacing capability.

Role-in-System

Ensure reliable communication between the microcontroller and connected components.

13. Oscillator Circuit (OSC)

Description

The oscillator circuit generates clock pulses required for microcontroller operation.

Function

- Provides timing signals.
- Synchronizes all internal operations.

Role-in-System

Controls the execution speed of instructions in the automatic door system.

14. Timing and Control Unit

Description

This unit controls overall timing and synchronization inside the microcontroller.

Function

- Generates control signals.
- Coordinates internal operations.

Role-in-System

Ensures smooth and synchronized operation of all system components.

15. Interrupt, Serial Port, and Timer Blocks

Description

These blocks provide advanced communication and timing features.

Function

- Timers generate delays.
- Serial port enables communication.
- Interrupts handle urgent events.

Role-in-System

Timers are used to generate delays for door opening and closing operations.

16. Control Pins (PSEN, ALE, EA, RST)

Description

These pins manage memory access and system control.

Function

- PSEN reads external memory.
- ALE separates address/data signals.
- EA selects memory type.
- RST resets the system.

Role-in-System

Provide proper initialization and memory control during operation.

Program

```
#include <reg51.h>

/* Motor pins */

sbit mot_pin1 = P1^1;

sbit mot_pin2 = P1^2;

sbit s=P1^0;

/* Delay function */

void Delay(int k);

void main(void)
{
    while(1)
    {
        // Clockwise rotation
        if(s==0){

            mot_pin1 = 1;

            mot_pin2 = 0;

            Delay(3000);

            // Stop

            mot_pin1 = 0;

            mot_pin2 = 0;

            Delay(3000);

            // Anticlockwise rotation
            mot_pin1 = 0;
```

```
mot_pin2 = 1;
Delay(3000);
    }
// Stop (optional, but cleaner cycle)

mot_pin1 = 0;

mot_pin2 = 0;

}
}
```

```
void Delay(int k)
{
    int i, j;
    for(i = 0; i < k; i++)
    {
        for(j = 0; j < 100; j++);
    }
}
```


3. IR Sensor

- **Function:**

Detects the presence or movement of a person using infrared radiation.

- **Role-in-Circuit:**

The IR sensor provides input to the microcontroller by detecting a person near the door. It triggers the system to open or close the door automatically.

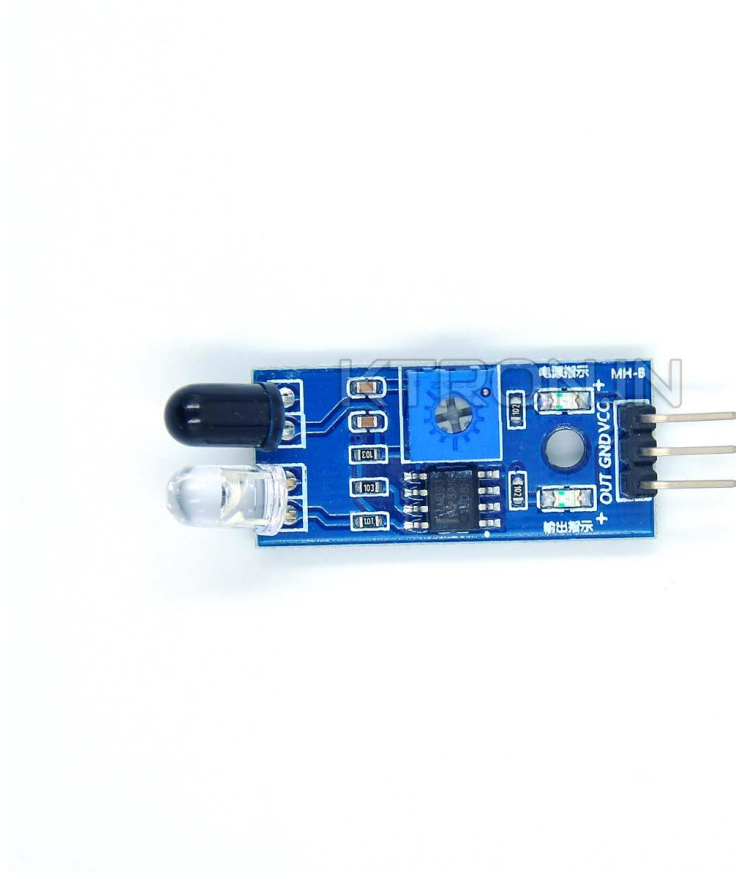


Fig.8 IR Sensor

1. VCC Pin

- Provides power supply to the IR sensor module.
- Usually connected to +5V supply.

2. GND Pin

- Ground pin of the sensor module.
- Connected to the negative terminal of the power supply.

3. OUT Pin

- Output pin of the IR sensor.
- Sends HIGH or LOW signal depending on object detection.
- Connected to the input pin of the microcontroller.

4. L293D Motor Driver IC

- **Function:**

Drives the DC motor by providing sufficient current and controlling direction.

- **Role-in-Circuit:**

The L293D acts as an interface between the microcontroller and the motor. It receives low-power signals from the microcontroller and converts them into high-power signals to drive the motor safely.



Fig.9 L293D Motor Driver IC

L293D IC Pins and Their Functions

1. Enable Pins (EN1, EN2)

- Used to enable the motor driver channels.
- Motor operates only when enable pins are HIGH.

2. Input Pins (IN1, IN2, IN3, IN4)

- Receive control signals from the microcontroller.
- Determine the direction of motor rotation.

3. Output Pins (OUT1, OUT2, OUT3, OUT4)

- Connected to the DC motor terminals.
- Provide output current to drive the motor.

4. VCC1 Pin

- Power supply pin for internal logic circuit.
- Usually connected to +5V.

5. VCC2 Pin

- Power supply pin for motor operation.
- Connected to motor supply voltage such as 9V or 12V.

6. GND Pins

- Ground pins of the IC.
- Connected to the common ground of the circuit

Function of L293D IC

- Drives DC motors using low-power control signals.
- Controls motor direction and operation.
- Provides current amplification.
- Protects the microcontroller from motor back EMF.

5. DC Motor

- **Function:**

Converts electrical energy into mechanical motion.

- **Role-in-Circuit:**

The DC motor is responsible for physically opening and closing the door. Its direction of rotation determines whether the door opens or closes.

- **Working Principle:**

Operates on the principle of electromagnetic force between magnetic fields and current-carrying conductors.

- **Direction Control:**

The direction of rotation changes by reversing the polarity of the supply voltage.

- **Mechanical Movement:**

Produces rotational motion that can be connected to gears, wheels, or door mechanisms.

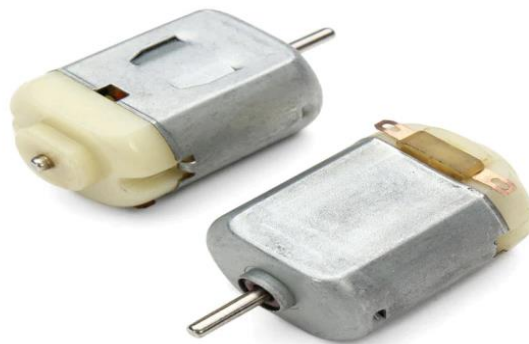


Fig.10 DC Motor

6. Crystal Oscillator

- **Function:**

Generates clock pulses required for microcontroller operation.

- **Role-in-Circuit:**

The crystal oscillator provides a stable clock signal (typically 11.0592 MHz) to the microcontroller, enabling proper timing and execution of instructions.

Timing Control:

Controls the speed at which the microcontroller executes instructions.

- **Frequency Stability:**

Maintains precise frequency with very little variation due to temperature or voltage changes.

- **Instruction Execution:**

Each instruction inside the microcontroller is executed according to the oscillator clock timing.

- **Serial Communication Support:**

Frequencies like 11.0592 MHz are preferred because they generate accurate baud rates for UART/serial communication.

- **Clock Pulse Generation:**

Produces periodic square-wave signals that act as synchronization pulses.

- **System Synchronization:**

Ensures all peripherals and internal modules operate in coordination.



. Fig.11 Crystal Oscillator

2. CONCLUSION

3.1 Conclusion

The automatic door opening system using the 8051 microcontroller has been successfully designed and implemented. This project demonstrates how embedded systems can be effectively used to automate real-world applications by integrating sensors, control units, and actuators.

The system efficiently detects the presence of a person using an IR sensor and performs the required action of opening and closing the door automatically. The 8051 microcontroller processes the input signals and controls the motor through the L293D motor driver, ensuring smooth and reliable operation. The use of proper timing and control logic helps in achieving accurate and stable performance.

This project highlights the importance of automation in improving convenience, safety, and hygiene, especially in public places such as hospitals, offices, and commercial buildings. It also reduces human effort and enhances system efficiency.

Furthermore, the project provides valuable practical knowledge in microcontroller programming, hardware interfacing, and system design. It serves as a strong foundation for developing more advanced automation systems.

In future, the system can be enhanced by integrating advanced technologies such as motion sensors, RFID-based access control, or IOT-based remote monitoring, making it more intelligent and secure.

Overall, the project successfully achieves its objectives and demonstrates a reliable and cost-effective solution for automatic door control.

3.1Future Scope

The automatic door opening system using the 8051 microcontroller can be further enhanced and upgraded with advanced technologies to improve its functionality, efficiency, and security. The future scope of this project includes the following:

- The system can be upgraded by replacing the IR sensor with advanced motion sensors or ultrasonic sensors for more accurate and reliable detection.
- It can be integrated with RFID (Radio Frequency Identification) technology to allow access only to authorized persons, improving security.
- The system can be connected to IOT (Internet of Things) platforms, enabling remote monitoring and control of the door through smartphones or computers.
- Facial recognition or biometric systems can be added for high-level security applications in offices and restricted areas.
- A display unit (LCD) can be incorporated to show system status such as “Door Open” or “Door Closed”.
- Voice control features can be integrated to operate the door using voice commands.
- Solar power supply can be used to make the system energy-efficient and suitable for remote locations.
- Safety features such as obstacle detection can be added to prevent accidents during door closing.
- The system can be implemented in smart home automation systems and integrated with other devices for centralized control.
- Wireless communication technologies like Bluetooth or Wi-Fi can be added for better flexibility and control.

Overall, the project has a wide scope for improvement and can be transformed into a smart, secure, and fully automated door control system suitable for modern applications.

4. BIBLIOGRAPHY

Bibliography

The following references and resources were used to complete the Automatic Door Opening System project, including textbooks, online materials, and research-based learning sources. These references helped in understanding the working of the 8051 microcontroller, IR sensors, motor driver circuits, and embedded system design.

Books & Text References

1. Muhammad Ali Mazidi — *The 8051 Microcontroller and Embedded Systems*, Pearson Education.
2. Kenneth J. Ayala — *The 8051 Microcontroller Architecture, Programming and Applications*, Penram International Publishing.
3. Boylestad & Nashelsky — *Electronic Devices and Circuit Theory*, Pearson Education.
4. Malvino & Leach — *Digital Principles and Applications*, McGraw Hill.
5. M.M. Mano — *Fundamentals of Digital Electronics and Microprocessors*, Prentice Hall India.

Web References

1. Electronics Tutorials – <https://www.electronics-tutorials.ws>
2. Circuit Digest – <https://circuitdigest.com>
3. Engineers Garage – <https://www.engineersgarage.com>
4. Electronic Hub – <https://www.electronicshub.org>
5. Elprocus Electronics – <https://www.elprocus.com>